

One data centre. Three connected plants.

J.B. Chemicals & Pharmaceuticals Ltd.

Turnkey point-to-point connectivity across Panoli and Ankleshwar: wireless engineering, tower infrastructure, network integration and deployment.

3

Plant locations

2

Wireless paths

P2P

Dedicated radio links

Panoli hub | Ankleshwar plant | Panoli Unit 2

The requirement | Bring three locations together

A shared technology backbone for distributed plants

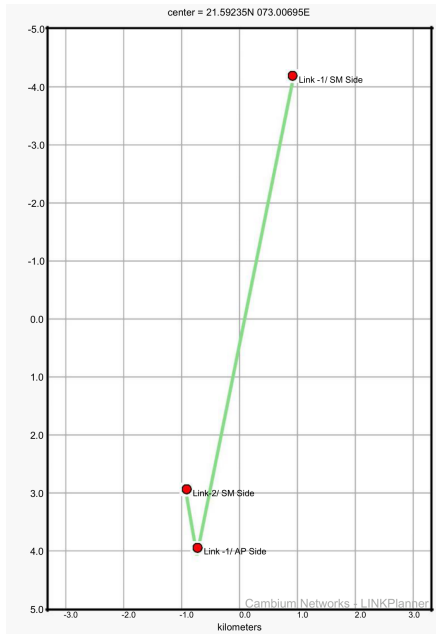
JB required connectivity between its Panoli data-centre location, the Ankleshwar plant and a second Panoli unit. D-TECH describes the engagement as delivered inter-plant connectivity: a central hub serving the local plant network and two remote plant locations over dedicated point-to-point wireless paths.

Delivery layer	D-TECH solution scope
Wireless engineering	Path profiles, antenna selection and capacity modelling using Cambium LINKPlanner.
Physical infrastructure	Tower, foundation, installation, earthing and aviation-light scope in the June 2018 order.
Network equipment	Cambium PTP 550 5 GHz radios, CAT6 STP cable and an Aruba 8-port Gigabit L2 switch in the November 2020 order.
Deployment & integration	Mount and align radio endpoints; connect the wireless hand-off to the client LAN and data-centre network.
Security & handover	Apply the approved network-access controls; validate connectivity and document operating settings.

Inside: connectivity design - page 3 | technology and security approach - page 4 | deployment photographs and client value - page 5

The proposed design | Two engineered wireless paths

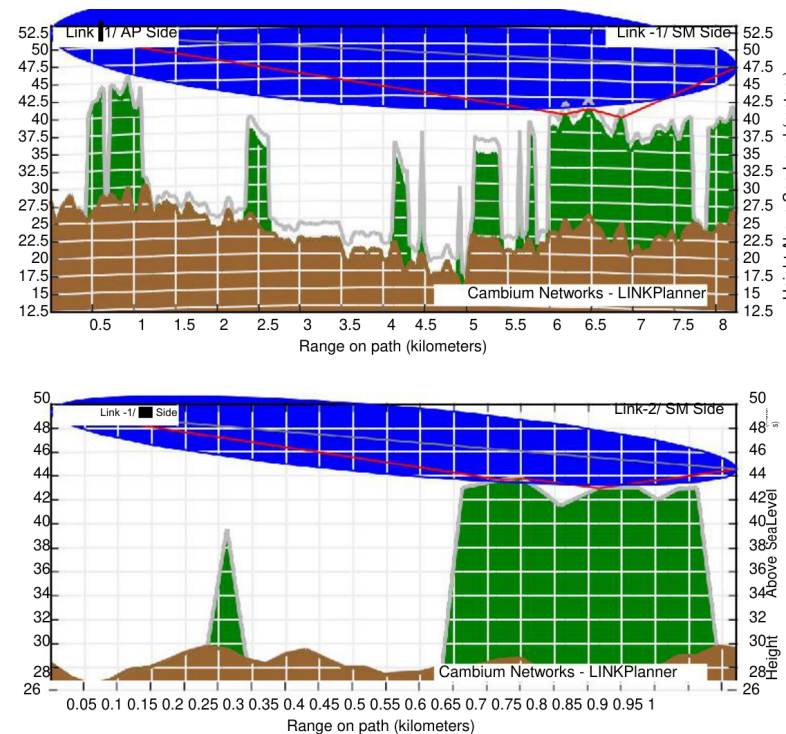
Panoli hub + two paths.



PROPOSED NETWORK MAP

Proposed operation: central data-centre LAN > Panoli hub > dedicated radio paths > remote plant LANs.

Map and profiles: supplied PTP 550 LINKPlanner proposal, pp. 4, 6 and 9 (February 2018). Heights are antenna heights above local ground. Route names follow D-TECH's description. Profiles model terrain, obstacles and Fresnel clearance; these are proposal settings, not verified as-built measurements.



PANOLI - ANKLESHWAR

Long path: 8.248 km
Connectorized PTP 550 radios
29 dBi dual-polar dish antennas
Modelled heights: 24 m / 21 m
5.8 GHz band; 40 MHz channel

PANOLI - PANOLI UNIT 2

Short path: 1.020 km
Integrated PTP 550 radios / antennas
Modelled heights: 21 m / 15 m
5.8 GHz band; 40 MHz channel

The technology | Engineered paths, controlled access

Planning item	Long path	Short path
Modelled distance	8.248 km	1.020 km
PTP 550 proposal: aggregate IP capacity	211.72 Mbps maximum in project summary	215.03 Mbps maximum in project summary
PTP 670 alternative: aggregate IP capacity	282.34 Mbps maximum in project summary	356.27 Mbps maximum in project summary
Antenna approach in the proposals	Connectorized radio + 29 dBi dual-polar dish	Integrated radio / antenna

PTP 550 is specified in the supplied 2020 equipment PO. PTP 670 is an alternative design from the 2018 report. All capacities above are modelled aggregate values, not measured throughput or per-direction guarantees.

Secure connectivity approach

Security combines controlled radio management with the client’s network policy: restrict administrative access, separate management and plant traffic where approved, and permit only authorised services through the network boundary. Tower earthing and surge protection address the physical exposure of outdoor equipment.

Implementation detail: the supplied records do not establish enabled encryption, firewall rules or final VLAN configuration. Confirm these controls in the commissioning and handover record before making a certified security claim.

The delivery | Field execution to connected operations



Client-supplied photographs: tower erection and rooftop installation activity.

Survey & design > Tower & cabling > Mount & align > Integrate & test > Handover

A turnkey delivery approach brings wireless design, supporting infrastructure and LAN integration into one coordinated engagement. The result described by D-TECH is a shared connectivity backbone across three plant locations, supporting access to central IT resources without a new inter-site cable route.

Project sources: LINKPlanner PTP 550 and PTP 670 proposals (February 2018); tower PO 4600066881 (15 June 2018); equipment PO 4700014893 (20 November 2020); client-supplied site images. Delivery narrative: D-TECH. Operational uptime and security test results are not quantified here.

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D-TECH | JB Pharma | Plant connectivity case study | 04 October 2026